

Investigating Real-Time Communication Between Asymmetric Processors with OpenAMP

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Abstract—Modern mixed-criticality embedded systems increasingly rely on Asymmetric Multiprocessing (AMP) architectures, coupling high-throughput application processors with deterministic real-time processing units. While OpenAMP traditionally provides Inter-Processor Communication (IPC) via the RPMsg protocol, this work investigates an emerging paradigm: *hypervisorless VirtIO* (AMP VirtIO). By substituting heavyweight hypervisor traps (VM-exits) with hardware Inter-Processor Interrupts (IPIs) and deploying a user-space Physical Machine Monitor (PMM), standard VirtIO backends (such as *virt-net*, *virt-rng*) are exposed directly to the Zephyr RTOS on a remote bare-metal core. Through architectural mapping, source-code analysis, and memory inspection within an emulated Xilinx Zynq UltraScale+ MPSoC, this work evaluates the end-to-end execution flow, profiles transport latency, and discusses architectural trade-offs against classic RPMsg. Furthermore, this study documents the porting effort onto physical silicon using a Xilinx ZCU102 board, evaluating microarchitectural discrepancies between emulation and physical deployment, and contextualizing the findings within the emerging *virtio-msg* specification.

Index Terms—Asymmetric Multiprocessing (AMP), Inter-Processor Communication (IPC), Mixed-Criticality Systems, OpenAMP, Real-Time Systems, VirtIO.

I. INTRODUCTION

Heterogeneous Asymmetric Multiprocessing (AMP) Systems-on-Chip (SoCs), such as the Xilinx Zynq UltraScale+ MPSoC, have become central to modern mixed-criticality designs. These platforms integrate high-performance Application Processing Units (APUs, e.g., ARM Cortex-A53) running general-purpose operating systems like Linux, alongside deterministic Real-time Processing Units (RPU, e.g., ARM Cortex-R5) executing a Real-Time Operating System (RTOS).

Inter-Processor Communication (IPC) across these heterogeneous execution domains is traditionally managed by the OpenAMP framework [1] via the Remote Processor Messaging (RPMsg) protocol [3] over shared memory. While RPMsg provides an established, low-overhead transport layer, it operates strictly on unformatted byte arrays. Consequently, sharing complex virtualized resources—such as network adapters or cryptographic hardware—requires developers to design proprietary Remote Procedure Call (RPC) layers.

The VirtIO standard [2], [11] addresses this limitation by offering standardized device abstraction models (e.g., *virtio-net*, *virtio-rng*). However, traditional VirtIO implementations are tightly coupled with Type-1 or Type-2 hypervisors (such as KVM or Xen), which intercept Memory-Mapped I/O (MMIO)

accesses via virtual machine exits (VM-exits). In hard real-time and safety-critical contexts, hypervisor context switching introduces non-deterministic scheduling jitter that degrades real-time guarantees.

To overcome this challenge, the **Hypervisorless VirtIO** (or AMP VirtIO) paradigm eliminates the hypervisor layer entirely. Instead, it utilizes a lightweight user-space Physical Machine Monitor (PMM) running on the Linux master, which maps VirtIO devices to fixed physical shared memory and substitutes hypervisor traps with hardware Inter-Processor Interrupts (IPIs).

This work conducts an architectural and experimental investigation of the hypervisorless VirtIO framework. Specifically, this work provides:

- An architectural formalization of the hypervisorless VirtIO execution model and memory topology;
- A code-level walkthrough of the Zephyr RTOS [6] firmware adaptations, detailing custom linker scripts, IPI driver implementations, and bounce-buffer shim layers;
- Static and dynamic physical memory validation within an emulated environment to verify descriptor structures and custom payload transfers;
- An evaluation of the physical hardware deployment on a Xilinx Zynq UltraScale+ ZCU102 MPSoC, including a root-cause analysis of ARM64 microarchitectural alignment faults;
- An approximate evaluation of Round-Trip Time (RTT) latencies within the emulated platform, alongside a qualitative architectural comparison with RPMsg and an alignment with emerging initiatives such as Linaro HVAC and the *virtio-msg* protocol.

II. SYSTEM ARCHITECTURE AND EXECUTION FLOW

To evaluate the operational mechanics of the hypervisorless VirtIO model, this study analyzes the reference deployment (*demo4*), which runs in a containerized environment emulating a heterogeneous MPSoC via QEMU. This section examines the hardware partitioning, software domains, and shared memory design that make hypervisor-free device virtualization possible.

A. Top-Level Architecture

The platform models an Asymmetric Multiprocessing (AMP) topology based on the Xilinx Zynq UltraScale+ MPSoC. As depicted in Fig. 1, the execution environment is divided into two distinct operational domains:

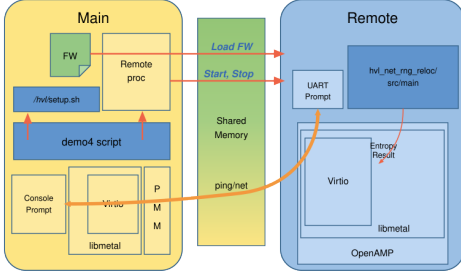


Fig. 1. Overview of the system architecture

- **Master (Host) Domain:** Deployed on an ARM Cortex-A53 Application Processing Unit (APU) cluster running a standard Linux kernel.
- **Remote (Real-Time) Domain:** Hosted on an ARM Cortex-R5F Real-time Processing Unit (RPU) executing the Zephyr RTOS, which runs the dedicated `hvl_net_rng_reloc` application.

Conventional VirtIO architectures rely on a heavyweight hypervisor such as KVM [14] to manage virtual machines and trap register manipulations through virtual machine exits (VM-exits). In contrast, this design operates completely without a hypervisor.

On the Linux host, a user-space Physical Machine Monitor (PMM)—running as an `lkvm` process [9]—instantiates backend abstractions for standard VirtIO devices, namely an Ethernet interface (*virt-net*) and a hardware random number generator (*virt-rng*), exposing them directly to the Zephyr RTOS. Inter-Processor Communication (IPC) between Linux and the Zephyr RTOS is mediated by the OpenAMP framework alongside the `libmetal` hardware abstraction layer [10], operating over a dedicated shared memory window.

To enable direct cross-core device communication without hypervisor trapping, the platform implements the `virtio-mmio` (Memory-Mapped I/O) transport interface over a fixed physical address space rooted at base address `0x37000000`.

B. Shared Memory Topology and VirtQueue Layout

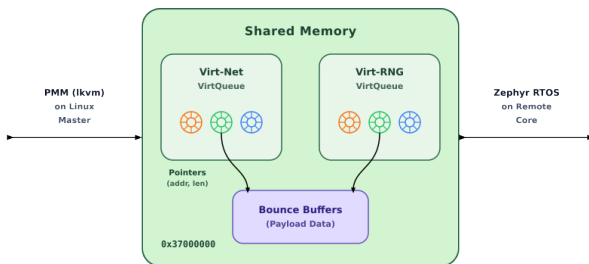


Fig. 2. Shared memory and VirtQueues detail

As illustrated in Fig. 2, both the Linux PMM and the remote Zephyr RTOS maintain symmetric read and write access to the shared memory window at base address `0x37000000`.

Within this memory space, two VirtIO backend devices are instantiated: *virt-net* for network traffic and *virt-rng* for entropy generation, each provisioning its own dedicated set of VirtQueues. To operate without a hypervisor, the architecture strictly decouples the control signaling from data payloads:

- **Control Plane:** Preserves the standard VirtIO three-ring layout (Descriptor, Available, and Used Rings).
- **Data Plane:** Replaces dynamic address resolution with pre-allocated, contiguous *bounce buffers* mapped directly in shared memory, ensuring the Descriptor Ring references mutually accessible physical addresses rather than private guest memory.

C. Component Description

To replace the role of a traditional hypervisor, the architecture relies on four key components:

Physical Machine Monitor (PMM): In an asymmetric multiprocessing system operating without hardware virtualization extensions, deploying a standard Virtual Machine Monitor (VMM) is conceptually inaccurate. Instead, the host runs a Physical Machine Monitor (PMM) based on a modified instance of `lkvm` from which all CPU virtualization features (e.g., KVM vCPU execution loops) have been removed. Operating purely in Linux user space, the PMM manages device lifecycle management and physical resource provisioning. It configures the shared memory base and exposes standard VirtIO backend devices, allowing the remote RTOS to interface with native device models without hypervisor mediation.

VirtIO-MMIO and Static Address Allocation: Communication between the cores relies on the standardized VirtIO Memory-Mapped I/O (`virtio-mmio`) transport. In contrast to conventional virtualized environments where memory spaces are allocated dynamically by guest drivers at runtime, this hypervisorless implementation binds devices to fixed, compile-time physical addresses. The PMM maps the configuration spaces and ring buffers for both *virt-net* and *virt-rng* into the shared memory segment starting at physical base address `0x37000000`, situated immediately after a 4 KB reserved alignment header.

Shared-Memory Bounce Buffers: Because both operating systems execute in disjoint address spaces without dynamic cross-domain address translation, pointers cannot be shared directly between them. To bridge this gap, a software shim layer routes all payload data through pre-allocated, fixed-size bounce buffers in shared physical RAM. Although this design introduces memory-copy (`memcpy`) overhead, it guarantees that descriptors reference valid physical addresses on both cores, eliminating invalid memory faults and race conditions between Linux and Zephyr.

Hardware Notifications (IPIs): Without hypervisor mediation to trap control-register accesses via VM-exits, the system relies on physical Inter-Processor Interrupts (IPIs) for asynchronous doorbell notifications. When either the Zephyr

driver or the Linux backend updates a VirtQueue or places new data into a bounce buffer, it issues a hardware IPI to notify the receiving core. On the Linux side, these inbound hardware interrupts are captured by a dedicated kernel module, `user-mbox.ko`, which services the interrupt and signals the user-space PMM process to consume pending descriptors.

D. Execution Flow

The end-to-end execution flow—from initial setup to bidirectional network interaction—is orchestrated primarily by the Linux host, as illustrated in Fig. 3.

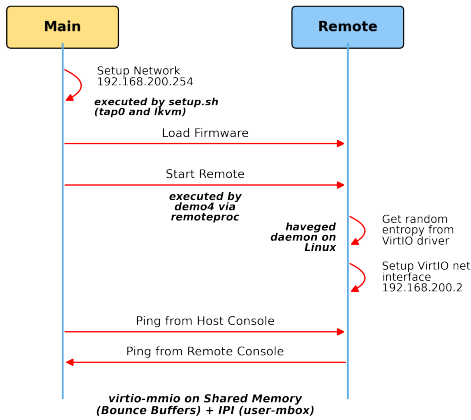


Fig. 3. Control flow diagram of the communication process

The initialization sequence begins by running the `/hvl/setup.sh` script on the Linux master. This script operates in the background to configure the virtual network interface (`tap0` at IP 192.168.200.254) and launch the `lkvm` PMM process. Concurrently, the `haveged` entropy daemon starts accumulating random data in user space for upcoming device queries.

Once the host infrastructure is active, the `demo4` script leverages the Linux `remoteproc` framework [12] to load the Zephyr firmware binary into memory and release the Cortex-R5F core from reset. Upon booting, the Zephyr RTOS initializes its `virt-rng` frontend driver and issues an initial request to pull the accumulated entropy from the shared memory segment (Fig. 4).

<pre> 0 "Main UART, Quad A53, Ctrl-A x to quit" 'mode=tap,tapif=tap0,trans=mmio' --vproxy Info: (virtio/mmio.c) virtio_mmio_init:620: virtio-mmio.de vices=0x200000x37000000 [0x4d564b4c:0x4] [30.592616] IPv6: ADDRCONF(NETDEV_CHANGE): tap0: link bec omes ready Info: (virtio/mmio.c) virtio_mmio_init:620: virtio-mmio.de vices=0x200000x37000000 [0x4d564b4c:0x1] we need to wait for the entropy to build up in the Linux sys tem Now we will start the firmware [37.041918] remoteproc remoteproc0: powering up r500 [37.065575] remoteproc remoteproc0: Booting fw image zeph yr.elf, size 2293856 [37.144264] r500: RPU boot from TCM. [37.158587] remoteproc remoteproc0: remote processor r500 is now up Look at the 2nd UART, you should see the results of the ent ropy test now we will ping the firmware through the virtio-net device PING 192.168.200.2 (192.168.200.2): 56 data bytes 64 bytes from 192.168.200.2: seq=0 ttl=64 time=20.310 ms 64 bytes from 192.168.200.2: seq=1 ttl=64 time=4.506 ms 64 bytes from 192.168.200.2: seq=2 ttl=64 time=4.476 ms --- 192.168.200.2 ping statistics --- 3 packets transmitted, 3 packets received, 0% packet loss round-trip min/avg/max = 4.476/9.764/20.310 ms now click in the 2nd UART pane to give it focus try these commands from the 2nd UART device list net iface net ping -c 20 192.168.200.254 root@generic-arm64:~# </pre>	<pre> 1 "dev@openamp-demo" This is a host shell After the QEMU board boots you can use ssh or scp from here Like \$ ssh qemu-zcu102 or \$ scp qemu-zcu102:my-results.log . If you exit this shell, it will exit the emulator also dev@openamp-demo:~/demo4-r5-hvl-virtio\$ 2 "2nd UART, R5_0" 0x3e 0x6d 0x3c 0x5f main:108 GET entropy test passed uart:-\$ 3 "3rd UART" waiting for port 1116 . Connecting </pre>
--	--

Fig. 4. Tmux session illustrating Zephyr boot sequence and entropy retrieval

Data-plane communication is subsequently validated through ICMP network traffic. When a ping command is executed from the Linux host console toward the Zephyr remote IP (192.168.200.2), the packets bypass physical network interfaces. Instead, payloads are written directly into the shared memory bounce buffers via the `virtio-mmio` transport layer, and an IPI is raised through the `user-mbox` driver to alert Zephyr. Finally, to demonstrate symmetrical bidirectional throughput, a reverse ping is initiated from the Zephyr console back to the Linux host (Fig. 5), completing the communication loop.

```

2 "2nd UART, R5_0"
uart:-$ net ping -c 20 192.168.200.254
net ping -c 20 192.168.200.254
PING 192.168.200.254
28 bytes from 192.168.200.254 to 192.168.200.2: icmp_seq=0 ttl=64 time=366 ms
28 bytes from 192.168.200.254 to 192.168.200.2: icmp_seq=1 ttl=64 time=176 ms
28 bytes from 192.168.200.254 to 192.168.200.2: icmp_seq=2 ttl=64 time=188 ms
28 bytes from 192.168.200.254 to 192.168.200.2: icmp_seq=3 ttl=64 time=256 ms
28 bytes from 192.168.200.254 to 192.168.200.2: icmp_seq=4 ttl=64 time=250 ms
28 bytes from 192.168.200.254 to 192.168.200.2: icmp_seq=5 ttl=64 time=220 ms
28 bytes from 192.168.200.254 to 192.168.200.2: icmp_seq=6 ttl=64 time=270 ms
28 bytes from 192.168.200.254 to 192.168.200.2: icmp_seq=7 ttl=64 time=246 ms
28 bytes from 192.168.200.254 to 192.168.200.2: icmp_seq=8 ttl=64 time=230 ms
28 bytes from 192.168.200.254 to 192.168.200.2: icmp_seq=9 ttl=64 time=234 ms
28 bytes from 192.168.200.254 to 192.168.200.2: icmp_seq=10 ttl=64 time=239 ms
28 bytes from 192.168.200.254 to 192.168.200.2: icmp_seq=11 ttl=64 time=220 ms
28 bytes from 192.168.200.254 to 192.168.200.2: icmp_seq=12 ttl=64 time=244 ms
28 bytes from 192.168.200.254 to 192.168.200.2: icmp_seq=13 ttl=64 time=210 ms
28 bytes from 192.168.200.254 to 192.168.200.2: icmp_seq=14 ttl=64 time=237 ms
28 bytes from 192.168.200.254 to 192.168.200.2: icmp_seq=15 ttl=64 time=245 ms
28 bytes from 192.168.200.254 to 192.168.200.2: icmp_seq=16 ttl=64 time=281 ms
28 bytes from 192.168.200.254 to 192.168.200.2: icmp_seq=17 ttl=64 time=253 ms
28 bytes from 192.168.200.254 to 192.168.200.2: icmp_seq=18 ttl=64 time=252 ms
28 bytes from 192.168.200.254 to 192.168.200.2: icmp_seq=19 ttl=64 time=228 ms
uart:-$

```

Fig. 5. Executing a ping from the Zephyr remote back to the Linux host

III. CODE-LEVEL ANALYSIS

To examine how the theoretical hypervisorless concepts translate into concrete firmware artifacts, this section analyzes the C source code of the Zephyr RTOS application. The implementation is based on the `hvl_net_rng_reloc` sample located in the experimental `virtio-exp` branch of the `zephyr-openamp-staging` repository.

A. Build-Time Configuration and Memory Binding

In conventional virtualized environments, VirtIO devices are dynamically orchestrated by a hypervisor that provisions VirtQueues within guest RAM and arbitrates feature negotiation. In the hypervisorless architecture, dynamic allocation must be bypassed entirely in favor of static physical shared-memory mappings.

This compile-time binding is established through specific configuration options in the Zephyr configuration file:

- `CONFIG_VIRTIO_MMIO=y`: Enables the VirtIO Memory-Mapped I/O transport subsystem within the RTOS.
- `CONFIG_COMPILER_OPT="-DHVL_VIRTIO"`: Defines a preprocessor macro that acts as a switch across the OpenAMP and `libmetal` libraries, routing calls to hypervisorless-specific routines rather than standard virtualized handlers.
- `CONFIG_HAVE_CUSTOM_LINKER_SCRIPT=y`: Instructs the build system to replace the default board memory map with an application-specific linker layout.

Because the system operates without dynamic address translation at runtime, the firmware requires static knowledge of the physical memory layout. This layout is defined in the custom linker script, `linker_r5_hvl.ld`.

The linker script explicitly reserves the physical shared memory boundary, defining a memory section named `shmem0` anchored at base address `0x37000000` (Fig. 6). This definition matches the base parameters initialized by the `lkvm` PMM during the Linux host bootstrap phase.

```

13 #define SHMEM_START_ADDR 0x37000000
14 #define SRAM2_ADDR      0x38000000
15 #define SRAM2_SIZE      0x00100000
16
17 #define SHMEM_RESERVED_HEADERS 4K
18 #define SHMEM_RESERVED_SIZE (16384 * 1K)
19
20 MEMORY
21 {
22     shmem0 (rw) : ORIGIN = (SHMEM_START_ADDR + SHMEM_RESERVED_HEADERS),
23                 LENGTH = (SHMEM_RESERVED_SIZE - SHMEM_RESERVED_HEADERS)
24
25     SRAM2 (woc) : ORIGIN = (SRAM2_ADDR), LENGTH = SRAM2_SIZE
26 }

```

Fig. 6. Section of `linker_r5_hvl.ld` defining the `shmem0` physical boundaries

Within this reserved region, the linker script forces the allocation of the VirtQueue control structures (Fig. 7):

```

28 SECTIONS
29 {
30     shmem0 (NOLOAD) : {
31         shmem0_start = .;
32         KEEP(*(.shared.vring.*));
33         shmem0_pos1_start = .;
34         shmem0_end = shmem0_start + (SHMEM_RESERVED_SIZE - SHMEM_RESERVED_HEADERS);
35     } > shmem0

```

Fig. 7. Linker directive for the allocation of VirtQueues

Using the directive

```
KEEP ( * (.shared.vring.*) )
```

the linker ensures that the Descriptor, Available, and Used Rings are placed directly into `shmem0` rather than the Cortex-R5F core's private tightly-coupled memory or local SRAM (SRAM2). As a result, the control rings are symmetrically accessible to both the Zephyr drivers and the Linux host PMM.

B. Hardware Signaling: Inter-Processor Interrupts vs. VM-Exits

In a conventional virtualized deployment, VirtIO notifications (commonly referred to as “kicks”) rely on memory-mapped doorbell register accesses that are intercepted by a hypervisor via hardware-assisted CPU virtualization traps (VM-exits). In this bare-metal AMP architecture, such traps are physically impossible because both Linux and Zephyr execute directly on bare metal without an intermediate supervisor layer. Consequently, hardware-level Inter-Processor Interrupts (IPIs) replace hypervisor-mediated traps.

This signaling logic is implemented in the Zephyr firmware within `src/xlnx_ipi.c`, which serves as a dedicated driver for the Xilinx ZynqMP IPI block. When the Zephyr frontend updates the Available Ring with new descriptors, the OpenAMP `virtio-mmio` transport layer triggers `xlnx_ipi_notify()` (Fig. 8). This function uses the `WRITE32` macro to perform a direct 32-bit register write targeting `IPI_TRIG3`. This operation raises a physical interrupt line on the APU, which is trapped by the Linux `user-mbox.ko` kernel module to wake the user-space PMM.

```

static void xlnx_ipi_notify(const struct device *dev)
{
    WRITE32(dev, IPI_TRIG, (1 << 24));
}

```

Fig. 8. Implementation of `xlnx_ipi_notify()`, showing the direct MMIO write to trigger the hardware IPI

Once the Linux backend finishes processing the buffers and updates the Used Ring, it asserts a return IPI targeting the Cortex-R5F. Rather than relying on a hypervisor to inject a virtual CPU interrupt (vCPU IRQ), Zephyr handles this directly through a registered Interrupt Service Routine, `xlnx_ipi_isr()` (Fig. 9). Inside this ISR, `virtio_mmio_hvl_cb_run()` serves as a callback into the OpenAMP libraries, prompting the frontend driver to scan the Used Ring for completed transactions (e.g., returning ICMP Echo Replies).

```

static void xlnx_ipi_isr(const struct device *dev)
{
    uint32_t isr, imr;
    uint32_t i;

    isr = READ32(dev, IPI_ISR);
    imr = READ32(dev, IPI_IMR);

    for (i = 0 ; i < 32; i++)
    {
        if (imr & (1 << i)) /* masked */
            continue;
        if (!(isr & (1 << i))) /* not active */
            continue;
        /* ACK interrupt */
        WRITE32(dev, IPI_ISR, (1 << i));
    }

#ifdef HVL_VIRTIO
    virtio_mmio_hvl_cb_run();
#endif
}

```

Fig. 9. Interrupt handler `xlnx_ipi_isr()` triggering the OpenAMP callback

C. Data-Plane Isolation and Bounce Buffers Implementation

In standard virtualization, memory access across the VirtIO boundary is dynamic: guest drivers write raw pointers referencing their local OS memory into the Descriptor Ring, and the hypervisor uses Stage-2 translation (Guest Physical Address to Host Virtual Address) to read the payload directly.

In this hypervisorless AMP setup, dynamic translation is unavailable. The Linux host running the PMM cannot directly or safely dereference the Cortex-R5F core’s private tightly-coupled memory or internal SRAM. Passing a local SRAM pointer across the boundary would trigger invalid memory faults on the host.

To bridge this architectural limitation without modifying standard upstream VirtIO frontend drivers, the transport layer (`virtio_mmio_hvl`) introduces a shim layer, which intercepts buffer management routines by overriding default VirtIO memory allocation functions through OS-dependent weak function bindings.

When the Zephyr network stack prepares an ICMP packet for transmission, the execution flow proceeds through three discrete stages:

- 1) **Shared Memory Allocation:** The shim intercepts the transmission request and allocates a contiguous memory block from the shared pool defined by the linker symbol `__shmem0_pool_start` within the `0x37000000` region.
- 2) **Explicit Copy (`memcpy`):** The payload bytes are copied from Zephyr’s private SRAM into the allocated shared bounce buffer using an explicit memory copy operation (abstracted via `libmetal` I/O block primitives).
- 3) **Descriptor Update:** Once the data is successfully copied, the driver updates the Descriptor Ring, setting the `addr` field to the physical address of the shared bounce buffer

(e.g., `0x37000xxx`) rather than the private SRAM pointer.

Finally, the driver triggers `xlnx_ipi_notify()`. When the host’s `user-mbox` catches the interrupt and signals the PMM, the backend reads the descriptor, resolves the physical pointer, and consumes the packet directly from the mutually accessible shared memory region. This architecture cleanly separates the control plane (the standard three-ring structures) from the data plane, enabling hypervisor-free operation at the expense of an explicit memory-to-memory copy overhead.

D. End-to-End Request Traces

A comprehensive tracking of a VirtIO request from start to finish, demonstrating how control and data signals traverse the hypervisorless boundary, is achieved by analyzing the execution paths of both an entropy read request (`virt-rng`) and an ICMP packet transmission (`virt-net`).

Virt-RNG Read Transaction: As illustrated in Fig. 10, execution initiates within the Zephyr runtime when `get_entropy()` is invoked. The OpenAMP transport shim allocates an empty bounce buffer inside the `shmem0` pool and appends its physical reference to the Descriptor Ring. Zephyr then triggers a hardware kick via `xlnx_ipi_notify()`. On the host domain, the PMM traps the interrupt, reads the available descriptor, fetches random bytes provided by the `havedged` daemon, and writes them directly into the shared bounce buffer. Finally, the PMM updates the Used Ring and fires a return IPI, which executes Zephyr’s `xlnx_ipi_isr()` and allows the driver to retrieve the entropy payload.

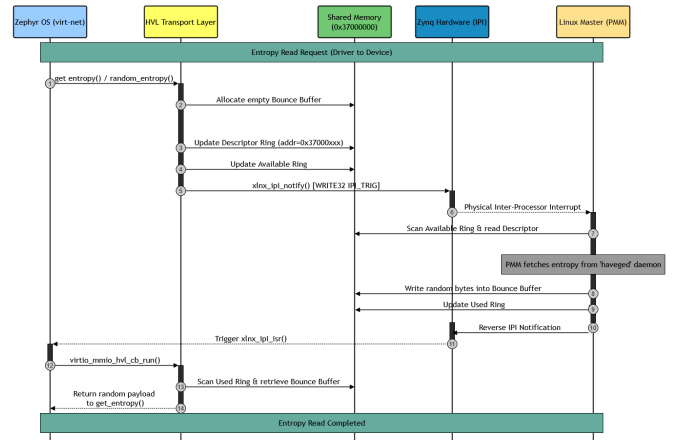


Fig. 10. Low-level sequence diagram of a `virt-rng` entropy read request.

Virt-Net Packet Transmission: As mapped in Fig. 11, outbound network frames follow an identical control flow but reverse the payload direction. When the Zephyr network stack generates an ICMP Echo Request, the transport layer intercepts the local buffer pointer, provisions a bounce buffer within the `0x37000000` boundary, and executes a synchronous memory copy (`memcpy`) of the packet bytes. After registering the physical buffer address in the Descriptor Ring, the firmware signals the host via the `IPI_TRIG` register. The Linux PMM

resolves the descriptor, reads the frame from shared memory, and injects it into the host's tap0 virtual interface.

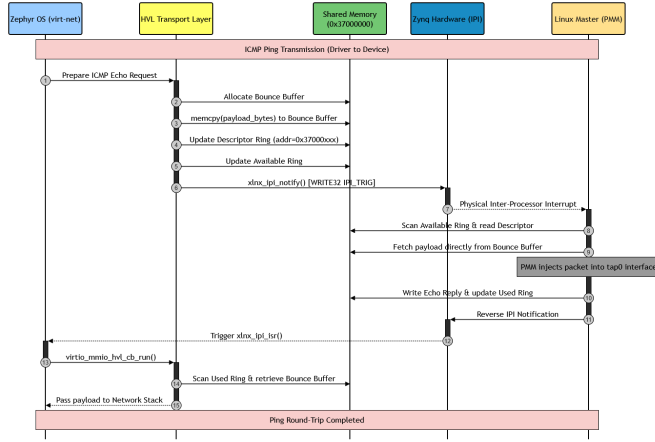


Fig. 11. Low-level sequence diagram of a virt-net ICMP ping transmission.

IV. EXPERIMENTS, MEMORY TRACING AND TIMING MEASUREMENTS

To validate the stability of the hypervisorless architecture beyond static code analysis, several experiments were conducted within the QEMU-emulated Zynq UltraScale+ environment with the goal of evaluating the data plane under increased workloads, inspecting memory structures, and verifying runtime data integrity via firmware instrumentation.

A. Data-Plane Stress Testing via Payload Modification

To verify the stability of the bounce-buffer layer under varying workloads, the network testing parameters in the demo4 suite were modified on the Linux host. The default ping routine was reconfigured to inject significantly larger ICMP data payloads using the `-s` flag (Fig. 12).



Fig. 12. ICMP ping transaction executed with a bigger payload

This alteration directly stresses the static bounce-buffer allocator within shared memory, forcing the OS-level shim

functions to allocate larger contiguous segments and perform extended memcopy operations. The Zephyr remote successfully serviced the incoming IPIs and responded to the enlarged echo requests without packet fragmentation or ring buffer overflows, demonstrating the functional robustness of the transparent shim layer.

B. Static Memory Layout Verification

On the Linux master, conventional user-space utilities (such as `devmem` or `hexdump`) were restricted by the kernel's `CONFIG_STRICT_DEVMEM` configuration, which blocks direct access to physical memory regions reserved by running device drivers. To bypass this restriction without altering kernel policies, physical memory was inspected out-of-band through the QEMU Monitor console using the physical examine command (`xp`).

```
(qemu) xp /32xw 0x37000000
0000000037000000: 0x74726976 0x00000001 0x00000004 0x4d564b4c
0000000037000010: 0x00000000 0x00000000 0x00000000 0x00000000
0000000037000020: 0x00000000 0x00000000 0x00001000 0x00000000
0000000037000030: 0x00000000 0x00000000 0x00000002 0x00001000
0000000037000040: 0xaabbaabb 0x00000000 0x00000000 0x00000000
0000000037000050: 0x00000000 0x00000000 0x00000000 0x00000000
0000000037000060: 0x00000000 0x00000000 0x00000000 0x00000000
0000000037000070: 0x00000004 0x00000000 0x00000000 0x00000000
(qemu)
```

Fig. 13. Physical memory dump at the shared memory base address

Decoding the 32-bit Little-Endian hexadecimal output at address `0x37000000` (Fig. 13) exposes the underlying VirtIO device configuration:

- `0x74726976`: Translates in ASCII to "virt", verifying the standard VirtIO MMIO identifier.
- `0x00000001`: Confirms compliance with the VirtIO Version 1 specification.
- `0x00000004`: Corresponds to Device ID 4, denoting a *virt-rng* (entropy generation) peripheral.
- `0x4d564b4c`: Decodes in ASCII to "LKVM", proving that the backend interface was established by the user-space `lkvm` daemon rather than a kernel hypervisor.
- `0xaabbaabb` (Offset `0x40`): Matches the compile-time acknowledgment macro `HVL_CFG_ACK` defined in Zephyr's `xlnx_ipi.c`.

This memory state confirms that the user-space PMM successfully injects native VirtIO MMIO device registers directly into physical shared memory.

C. Dynamic Payload Inspection via Descriptor Ring Traversal

To verify that network frames pass through the allocated bounce buffers without hypervisor mediation, a multi-step tracing was conducted to intercept an active payload in transit:

Step 1 – Custom Traffic Injection: On the Linux master, the background ping command was configured to flood the remote core with a known byte sequence. As the BusyBox suite only supports single-byte patterns, the hexadecimal byte `0x52`

```
ping -p 52 -s 128 -c 500 192.168.200.2 > /dev/null &
```

Step 3 – Descriptor Ring Inspection: Knowing from `linker_r5_hv1.ld` that the control rings immediately follow the 4 KB header reservation (anchored near `0x37001000`), the memory region was inspected using `xp /256xw` (Fig. 14). Inspecting the active descriptor entries revealed a physical buffer reference pointing to address `0x370078e8`.

```
root@generic-arm64:~# ping -p 52 -s 128 -c 500 192.168.200.2 > /dev/null &
root@generic-arm64:~# QEMU 5.1.0 monitor - type 'help' for more information
(qemu) xp /256kw 0x37001000
0000000037001000: 0x370078e3 0x00000000 0x000000b4 0x00010000
0000000037001010: 0x00000000 0x00000000 0x00000000 0x00020000
0000000037001020: 0x00000000 0x00000000 0x00000000 0x00030000
0000000037001030: 0x00000000 0x00000000 0x00000000 0x80000000
0000000037001040: 0x00029000 0x00000000 0x00000000 0x00000000
0000000037001050: 0x00000000 0x00000000 0x00000000 0x00000000
0000000037001060: 0x00000000 0x00000000 0x80000000 0x00000000
0000000037001070: 0x00000000 0x00000000 0x00000000 0x00000000
0000000037001080: 0x00000000 0x00000000 0x00000000 0x00000000
0000000037001090: 0x00000000 0x00000000 0x00000000 0x00000000
```

Step 4 – Payload Extraction: A targeted memory dump was executed at the resolved buffer pointer, formatted as raw ASCII characters using `xp /128c 0x370078e8` (Fig. 15). The resulting dump reveals the network encapsulation headers generated by *virt-net*, followed immediately by a contiguous block of cleartext 'R' characters.

[illegible]

D. Firmware Instrumentation and VirtIO-RNG Verification

In the baseline firmware, the entropy interface performs a sanity check during boot, requesting 10 bytes from the *virt-rng* device and verifying that the buffer is non-zero. To inspect the full data exchange, the Zephyr `get_entropy()` routine was modified to request an expanded 32-byte payload and dump the retrieved bytes directly to the auxiliary UART console.

```
2 "2nd UART, R5_0" -----
0x37
0x23
0xb3

=== CUSTOM INSTRUMENTATION RNG ===
Requested and received 32 entropy bytes from Linux PMM!
Hexadecimal values read by VirtIO-RNG:
0a 1f 9d 37 bc fd 20 ac
4b 43 ac 5f ee 48 b3 db
68 1b 48 5d 84 5d 00 ce
cb 9b 2e e3 37 23 b3 00
-----

main:121 - get_entropy test passed
```

This modification successfully triggered the full communication loop:

- Upon catching the interrupt, `xlhx_ipi_isr()` executed the OpenAMP callback, and the instrumented application dumped the matching 32-byte hexadecimal sequence to the UART interface (Fig. 16), confirming end-to-end data plane integrity.

To characterize the latency profile of the hypervisorless VirtIO transport, approximate Round-Trip Time (RTT) measurements were captured along the Inter-Processor Communication (IPC) path using bidirectional ICMP echo requests between the Linux master (APU) and the Zephyr remote (RPU).

- **Minimum Latency:** ~ 4.47 ms
- **Average Latency:** ~ 9.76 ms
- **Maximum Latency:** ~ 20.31 ms

Impact of Payload Scaling: To assess data-plane scaling, packet sizes were increased from the 56-byte default to

1400 bytes, approaching the standard Ethernet Maximum Transmission Unit (MTU). This modification elevated the maximum recorded latency to ~ 22.77 ms. The latency increase stems directly from the larger contiguous memory allocations in shared memory and the extended duration of the explicit `memcpy` operations required to stage and unstage 1400-byte buffers on both cores before interrupt signaling.

Emulation Artifacts vs. Industrial Real-Time Constraints:

The observed latencies should not be interpreted as the absolute real-time performance of the physical hardware. Because the experiments were conducted within a QEMU virtual platform running inside a Docker container on an Ubuntu host, both the Cortex-A53 and Cortex-R5F cores were scheduled as standard user-space threads. Consequently, the recorded measurements are heavily dominated by host operating system scheduling intervals, thread preemption, and container virtualization overhead.

These empirical trends confirm the functional integrity and copy overhead of the transport, but they cannot evaluate strict compliance with hard real-time deadlines. For instance, in automated electrical substation environments governed by IEC 61850 [7], critical Layer-2 control messages—such as Generic Object-Oriented Substation Events (GOOSE) and Sampled Values (SV)—demand deterministic end-to-end delivery latencies strictly below 3 ms. While host scheduling jitter in QEMU pushes latencies beyond this operational ceiling, evaluating true real-time determinism and worst-case execution time (WCET) requires deploying this stack onto physical silicon.

V. PHYSICAL DEPLOYMENT ON XILINX ZCU102: CHALLENGES AND ROOT-CAUSE ANALYSIS

To bridge the gap between containerized QEMU emulation and physical hardware execution, deployment of the hypervisorless VirtIO architecture was evaluated on a **Xilinx Zynq UltraScale+ MPSoC ZCU102** evaluation board [4]. The physical testbed comprised:

- **Host APU (Cortex-A53):** Running PetaLinux 2020.2 (Linux kernel 5.4, GLIBC 2.30), hosting the user-space `lkvm` PMM process.
- **Remote RPU (Cortex-R5F):** Running the Zephyr RTOS firmware payload, managed via Linux `remoteproc`.
- **Shared Memory:** A 16MB physical DDR aperture anchored at base address `0x37000000`.

While the real-time firmware was successfully built and staged, physical deployment exposed severe bottlenecks across the user-kernel boundary and the ARM64 memory subsystem. **The physical end-to-end communication remains currently unresolved** due to a fatal alignment fault on the host APU. The following subsections detail the firmware adaptations, the architectural root cause of the crash, and the evaluated mitigation strategies.

A. Firmware Adaptation

Porting the Zephyr firmware to the physical Cortex-R5F core required resolving a preliminary build-time challenge regarding the entropy subsystem. Specifically, compiling the firmware

with VirtIO support triggered an unresolved linker error for `zephyr_entropy`, caused by the absence of a physical True Random Number Generator (TRNG) node in the default Cortex-R5 devicetree. This was mitigated in `prj.conf` by substituting the missing hardware peripheral with a software pseudo-random generator:

```
CONFIG_TEST_RANDOM_GENERATOR=y
CONFIG_ENTROPY_GENERATOR=y
```

While this produced an operational ELF artifact (`zephyr.elf`) loadable by `remoteproc`, initialization stalled on the Linux host domain.

B. Memory Attribute Conflict and ARM64 Alignment Faults

While the QEMU reference environment relied on a custom staging kernel driver (`user-mbox.ko`) to expose the hardware IPI doorbell to user space, physical deployment on PetaLinux replaced this out-of-tree module with the mainline Linux Userspace I/O framework (`uio_pdrv_genirq` via `/dev/uio0`), binding directly to the platform IPI interrupt line without requiring custom kernel patches.

Upon starting `lkvm` on the physical ZCU102 target with shared memory parameters bound to `/dev/uio0`:

```
lkvm run --shmem-addr 0x37000000 --shmem-size 0x1000000 ...
```

the user-space process terminated abruptly with a `Bus error`.

Low-level syscall tracing via `strace` isolated the fault immediately after the `mmap()` invocation on the UIO file descriptor, specifically at offset `+0x80` within the mapped page:

```
openat(AT_FDCWD, "/dev/uio0", O_RDWR) = 3
mmap(NULL, 16777216, PROT_READ|PROT_WRITE,
      MAP_SHARED, 3, 0) = 0xffffb7000000
--- SIGBUS {si_signo=SIGBUS, si_code=BUS_ADRALN,
      si_addr=0xffffb70000080} ---
+++ killed by SIGBUS +++
```

Architectural investigation revealed that this fault is not a software logic error, but an intractable microarchitectural conflict between the kernel UIO driver [13] and the ARMv8-A memory model:

- **Kernel Page Attributes:** The platform's default UIO driver (`uio_pdrv_genirq`) is designed for memory-mapped peripheral registers. Consequently, it maps physical pages into user space using `pgprot_noncached()`, tagging them in the ARM64 Translation Table Descriptors (TTE) as *Device Memory* (`Device-nGnRE`).
- **ARMv8-A Microarchitecture Constraint:** Under the ARMv8 architecture [5], multi-register load/store operations—specifically the 64-bit paired store instruction (STP)—are **strictly prohibited** on regions tagged as *Device Memory*, irrespective of whether the physical target address is naturally aligned.
- **Compiler Optimization:** When `lkvm` initializes the VirtIO PMM header at offset `+0x80`, GCC emits an optimized paired-store instruction (`STP xzr, xzr, [x0, #128]`) to zero out the structure. The Cortex-A53 MMU hardware detects an STP instruction targeting a *Device Memory* page, immediately asserts a hardware Alignment

Fault (BUS_ADRALN), and terminates the process via SIGBUS.

C. Evaluated Mitigation Strategies and Unresolved Status

To obtain *Normal Non-Cacheable* page attributes on the shared aperture without rebuilding the platform Board Support Package (BSP), several user-space and kernel-level mitigations were explored (Table I).

TABLE I
DEPLOYMENT WORKAROUNDS AND FAILURE ROOT CAUSES ON ZCU102

Strategy	Layer	Outcome and Failure Cause
User-space Redirection to /dev/mem	User space	Failed: Direct mapping still enforced Device memory attributes (BUS_ADRALN) and severed the IPI signaling path.
Forced O_SYNC on /dev/mem	User space	Failed: Kernel memory reservations and security policies prevented Write-Combining promotion.
Dynamic Driver Re-bind (uio_dmem)	Kernel / Sysfs	Failed: The pre-compiled devicetree lacked a CMA-backed coherent memory reservation pool.

User-Space Shims (LD_PRELOAD): To bypass the restrictive attributes imposed by `uio_pdrv_genirq` without modifying the `lkvm` binary, a runtime shim utilizing `LD_PRELOAD` (supplemented by direct `svc #0` syscall interception) was developed to redirect `/dev/uio0` calls toward raw physical memory via `/dev/mem`. However, this strategy proved architecturally unviable on two fronts. First, under ARM64 Linux, mapping physical memory outside system RAM via `/dev/mem` still enforces *Device Memory* attributes; attempts to force Write-Combining via `O_SYNC` were blocked by kernel policies and `CONFIG_STRICT_DEVMEM`, triggering the identical `BUS_ADRALN` fault on paired stores (STP). Second, even if memory attributes had been promoted, `/dev/mem` operates strictly as a passive memory window without interrupt line bindings. Bypassing UIO completely severed the control plane, leaving the PMM unable to receive asynchronous IPI doorbells from the Cortex-R5F.

Kernel Driver Transition: Properly resolving this bottleneck requires replacing `uio_pdrv_genirq` with `uio_dmem_genirq`, which provisions physical buffers with *Normal Non-Cacheable* attributes via coherent DMA mappings. Dynamic rebinding via `sysfs` failed because the pre-compiled devicetree lacked a dedicated `reserved-memory` node managed by the Contiguous Memory Allocator (CMA).

Consequently, **hardware deployment remains an open challenge**. Eliminating the fault requires modifying the PetaLinux devicetree (DTS) to define a CMA coherent memory pool and rebuilding the kernel BSP with native `CONFIG_UIO_DMEM_GENIRQ` support, confirming that hypervisorless VirtIO on bare-metal ARM64 systems demands tightly coupled kernel memory management and devicetree co-design.

VI. SYNTHESIS AND ARCHITECTURAL TRADE-OFFS

A. VirtIO Protocol Adaptations

Operating the standard `virtio-mmio` transport on bare-metal hardware requires adapting several mechanisms that conventionally depend on hypervisor mediation:

- **Static Feature Resolution:** In hypervisor-managed deployments, device features are negotiated dynamically at boot via virtualized register reads and writes. In this hypervisorless implementation, dynamic negotiation is replaced by predefined feature masks declared at compile time and placed directly into the shared memory configuration header.
- **Transparent Buffer Redirection:** Standard VirtIO frontend drivers allocate `VirtQueues` and buffer spaces dynamically from guest-local RAM. Here, the memory allocation routines are overridden using OS-dependent weak function aliases, transparently redirecting all buffer requests to the statically configured `shmem0` pool without modifying upstream driver logic.

B. Comparative Analysis: VirtIO vs. RPMsg

Table II contrasts hypervisorless VirtIO against the conventional OpenAMP communication pipeline based on RPMsg.

TABLE II
ARCHITECTURAL TRADE-OFFS: HYPERVERSORLESS VIRTIO VS. RPMSG

Metric	Hypervisorless VirtIO	Classic RPMsg
Abstraction Level	High-level peripheral specifications (<i>net, mg</i>).	Low-level, unstructured byte-stream channels.
Software Reuse	High; uses standard upstream OS device drivers.	Low; requires custom application-level RPC protocols.
Data-Plane Zero-Copy	Unavailable; relies on intermediate bounce buffers.	Supported when endpoints share memory pools.
Memory Footprint	Fixed physical memory areas and hardcoded off-sets.	Configurable, dynamic or static ring topologies.
Control Signaling	Direct MMIO register writes driving physical IPIs.	Hardware mailboxes mapped to vrrings.

RPMsg provides a minimal, low-overhead transport suitable for passing raw byte arrays, but it lacks semantic awareness of the data it carries. Exposing complex host capabilities (such as network interfaces or storage blocks) over RPMsg requires developers to design, maintain, and serialize custom Remote Procedure Call (RPC) layers.

Hypervisorless VirtIO eliminates this fragmentation by elevating the interface to a standardized device specification. Because both the Linux kernel and the Zephyr RTOS include native VirtIO frontends, the remote core can consume host services as if they were physical hardware peripherals. The primary cost of this abstraction is reduced data-plane efficiency: while RPMsg can achieve true zero-copy performance, current hypervisorless VirtIO implementations incur the latency and memory overhead of intermediate bounce buffers.

C. Current Architectural Limitations

Memory Isolation and Fault Containment: Because the PMM operates as a user-space process on Linux with direct read/write mapping to the shared memory block (0x37000000), memory safety is a primary concern. In systems lacking hardware-enforced memory protection (such as an ARM TrustZone configuration, an RPU Memory Protection Unit, or an interconnect System MMU), a software fault or security compromise in the host domain can compromise the memory space of the real-time core.

Copy Overhead and Data Throughput: Standard VirtIO frontend drivers allocate descriptors pointing to local memory buffers. Because the real-time core cannot resolve dynamic host addresses, the transparent shim must execute an explicit `memcpy` into the shared bounce-buffer pool for every incoming and outgoing transaction. This extra memory copy increases CPU utilization on the real-time core and sets an upper bound on sustained network throughput.

Static Configuration: The current implementation depends heavily on hardcoded physical memory maps (e.g., anchoring `shmem0` at 0x37000000) defined within custom linker scripts. The architecture lacks a dynamic discovery mechanism, limiting runtime portability across different hardware platforms.

D. Real-Time Relevance: Core vs. Interconnect Determinism

Evaluating the real-time suitability of this architecture requires separating local processor execution from the cross-core communication path:

- **Local Real-Time Determinism:** Removing the hypervisor restores determinism to the Cortex-R5F core. Local RTOS tasks and native interrupt handlers execute on bare metal, completely insulated from hypervisor traps, timer emulation overhead, and virtualization jitter.
- **Cross-Domain IPC Determinism:** Inter-core communication remains non-deterministic. Because the backend device emulation resides in a user-space daemon on the Linux host, request completion times depend on host kernel scheduling, thread priorities, and interrupt processing in `user-mbox.ko`. Consequently, real-time control loops must run autonomously on the RTOS and avoid synchronous dependencies on the host VirtIO backend.

VII. FUTURE OUTLOOK: THE VIRTIO-MSG ROADMAP

The bounce-buffer mechanisms and static compile-time mappings used in this prototype represent an intermediate step toward standardized, hypervisor-free device virtualization.

This model is currently being formalized by Linaro's Heterogeneous VirtIO for Automotive Computing (HVAC) working group [8] through the development of the `virtio-msg` specification, specifically the `virtio-msg-amp` profile. The initiative focuses on three main objectives:

- Standardizing device configuration and ring-buffer semantics across non-cache-coherent, asymmetric processor topologies;

- Defining direct zero-copy buffer handoff protocols to eliminate intermediate bounce-buffer copies;
- Establishing a platform-agnostic signaling interface to decouple VirtIO doorbells from vendor-specific IPI hardware registers.

By addressing these architectural limitations, emerging standards aim to combine the code reuse of VirtIO device models with the determinism required for mixed-criticality industrial and automotive systems.

VIII. CONCLUSION

This work investigated the hypervisorless VirtIO framework as an alternative to classic RPSG for inter-processor communication across asymmetric processor cores. By anchoring `virtio-mmio` interfaces in static shared memory, substituting hypervisor VM-exits with hardware Inter-Processor Interrupts, and managing backend services via an unprivileged user-space Physical Machine Monitor, the architecture exposes standardized network and entropy peripherals to a bare-metal RTOS without hypervisor overhead.

Within the QEMU emulated platform, physical memory inspection and dynamic ring-buffer tracing verified that control plane signaling and bounce-buffer data transfers operate reliably across execution domains. While this paradigm successfully eliminates virtualization jitter on the real-time core, data-plane throughput remains bounded by intermediate memory copies and host-side scheduling variability.

Furthermore, hardware deployment on the Xilinx ZCU102 platform exposed critical physical constraints: generic user-space I/O drivers map shared memory with *Device Memory* attributes, triggering unrecoverable ARM64 hardware alignment faults on compiler-optimized paired stores (STP). The evaluated workarounds demonstrated that user-space shims cannot circumvent MMU page attributes without severing IPI interrupt delivery, proving that hypervisorless VirtIO on bare-metal ARMv8 platforms necessitates dedicated kernel DMA-coherent memory reservations. Addressing these microarchitectural memory constraints alongside emerging specifications, such as Linaro's `virtio-msg`, will be vital to establishing robust, zero-copy device virtualization for next-generation mixed-criticality systems.

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